

The Application of Systems Engineering in Project Management

—A Systems Engineering Analysis of TikTok Shop's GMV Growth &
Operational Systems in the Philippine Market

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Philippines Market

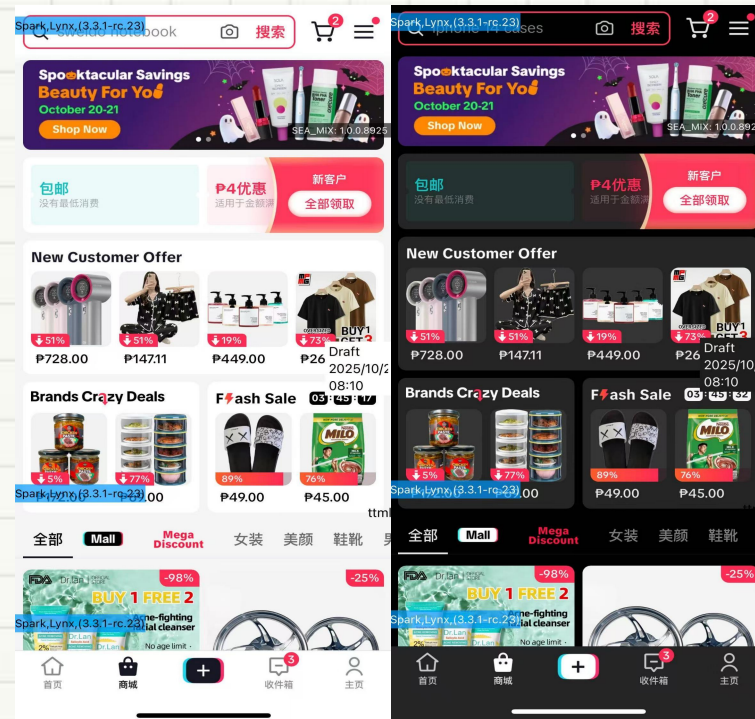
Strategic Operations

Shop Tab

TikTok Shop Philippines: GMV Growth Operations System

Business Background

Focusing on high-growth markets in Southeast Asia, strategic operations under the e-commerce model.



Market Characteristics

Extremely price-sensitive; average order value ~\$8–12 (USD).

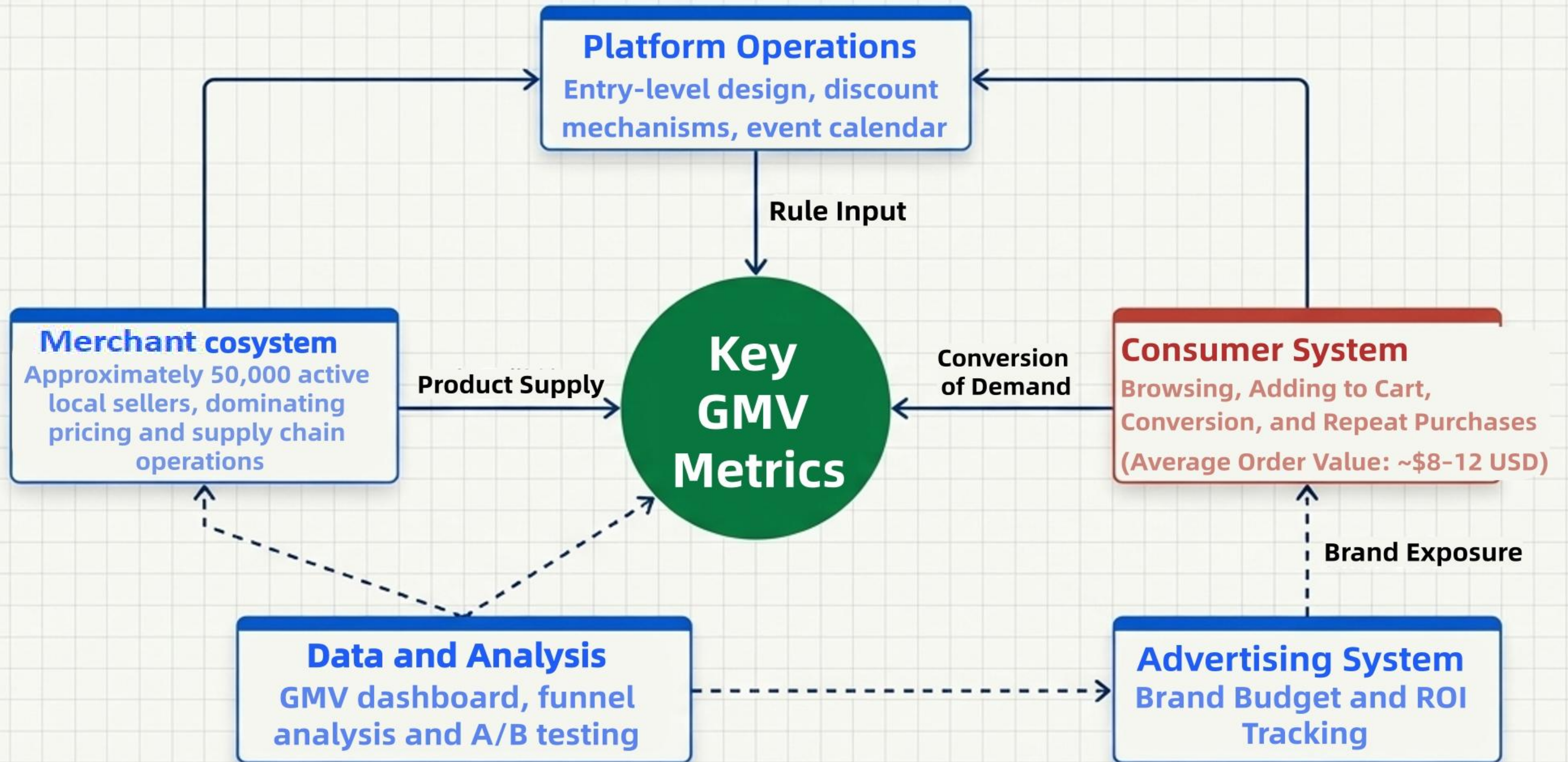
Key Metrics:
Achieving sustainable
GMV growth

Three Major Core Challenges

1. Coordination challenges across platform functions, merchant incentives, and consumer expectations.
2. Data latency limits real-time decision-making.
3. Local characteristics (unique payment habits, cultural cycles, and price sensitivity).



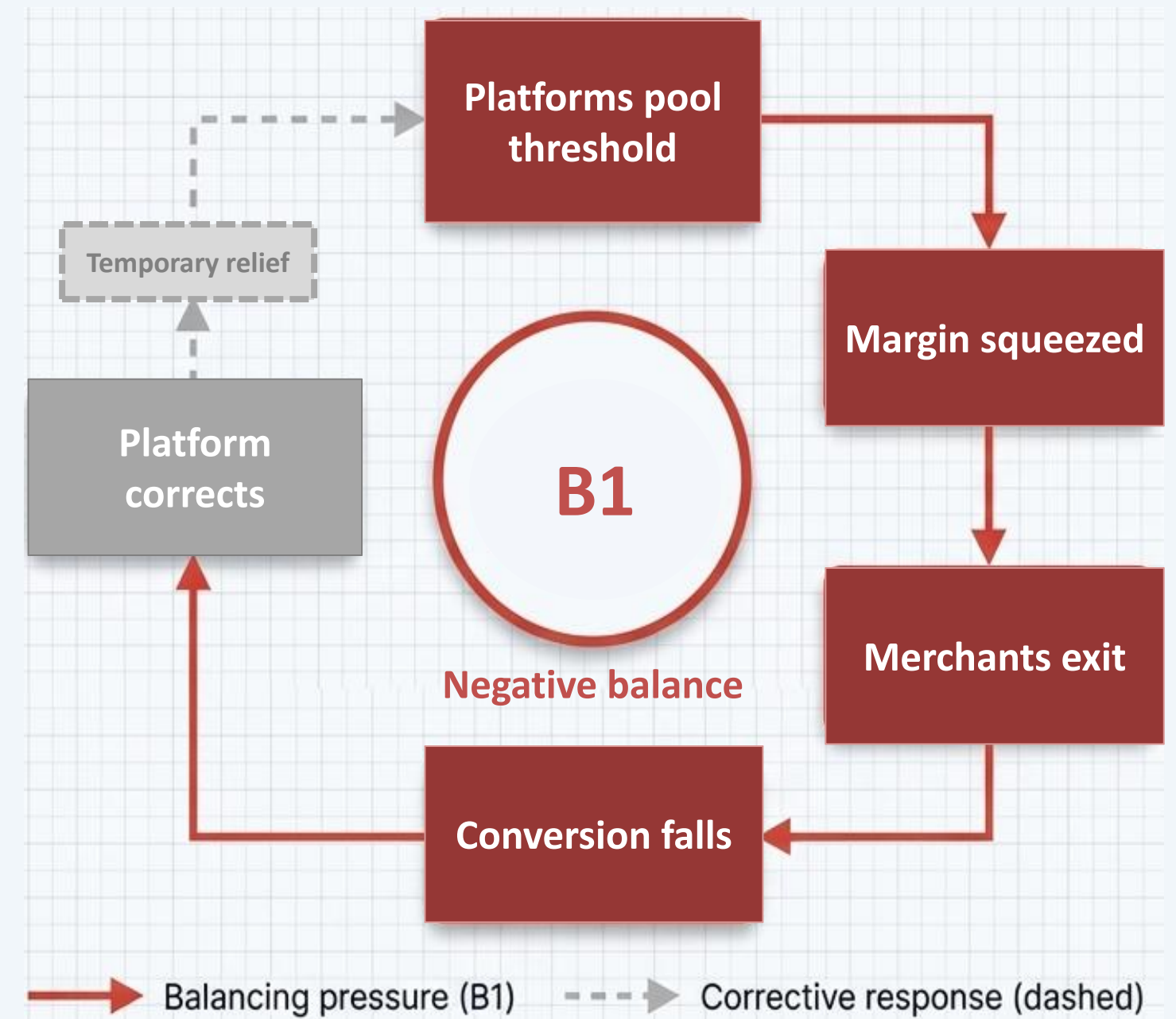
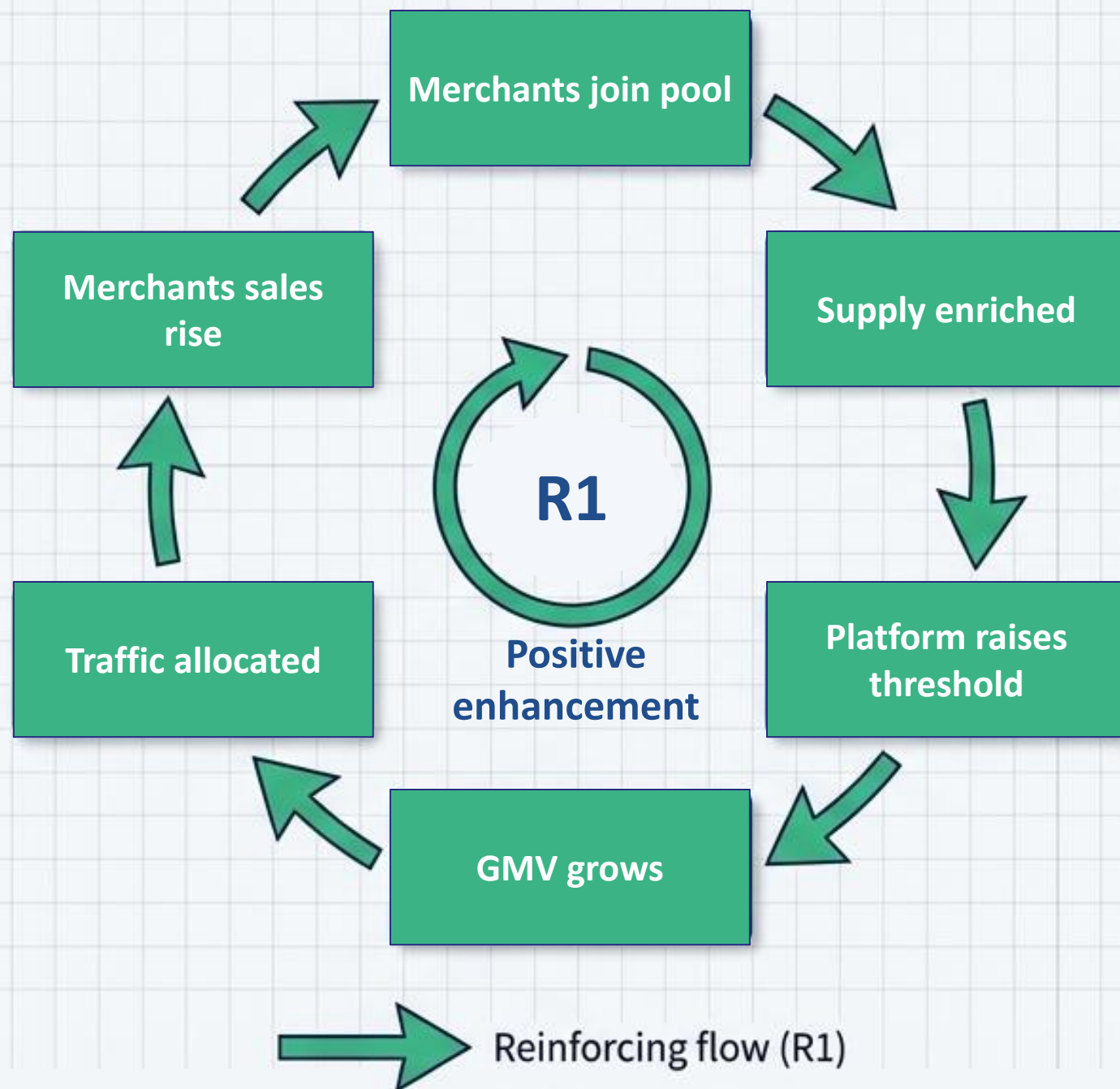
The Five Core Subsystems Driving Growth



[Solid line] Direct business flow · [Dotted line] Data and feedback flow
All subsystems are closely coupled through data, funds and product flows.

System Dynamics: R1/B1 Feedback Loop Dynamics

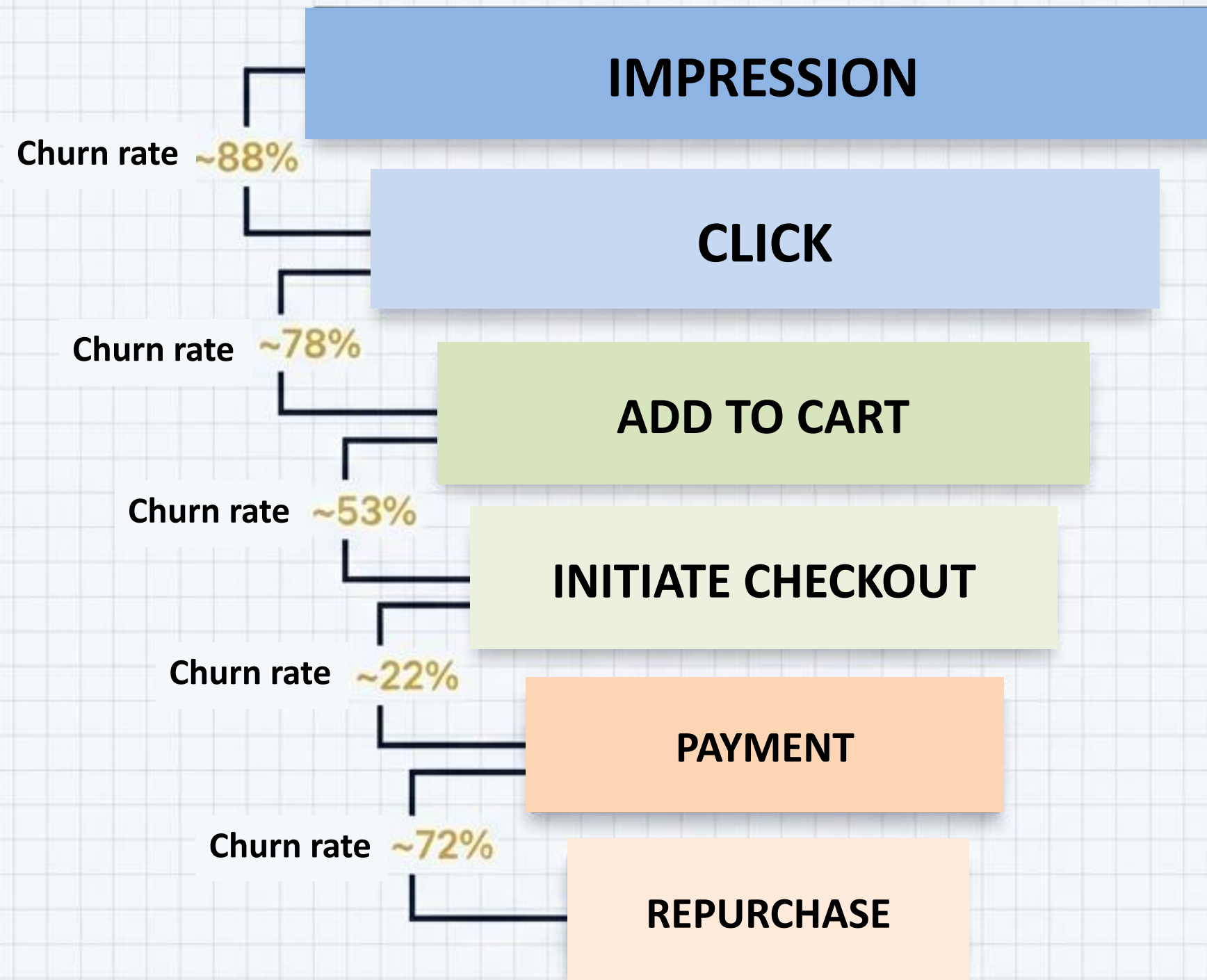
R1 Reinforcing Loop — Virtuous Growth Cycle ↻ B1 Balancing Loop — Over-threshold Trap ⊖



Key Insights: Actively managing merchant engagement—the addition of every high-quality merchant generates non-linear GMV growth.

Key Insight: Setting the bar too high will lead to merchant attrition, which will ultimately undermine the platform's efforts to grow its GMV targets.

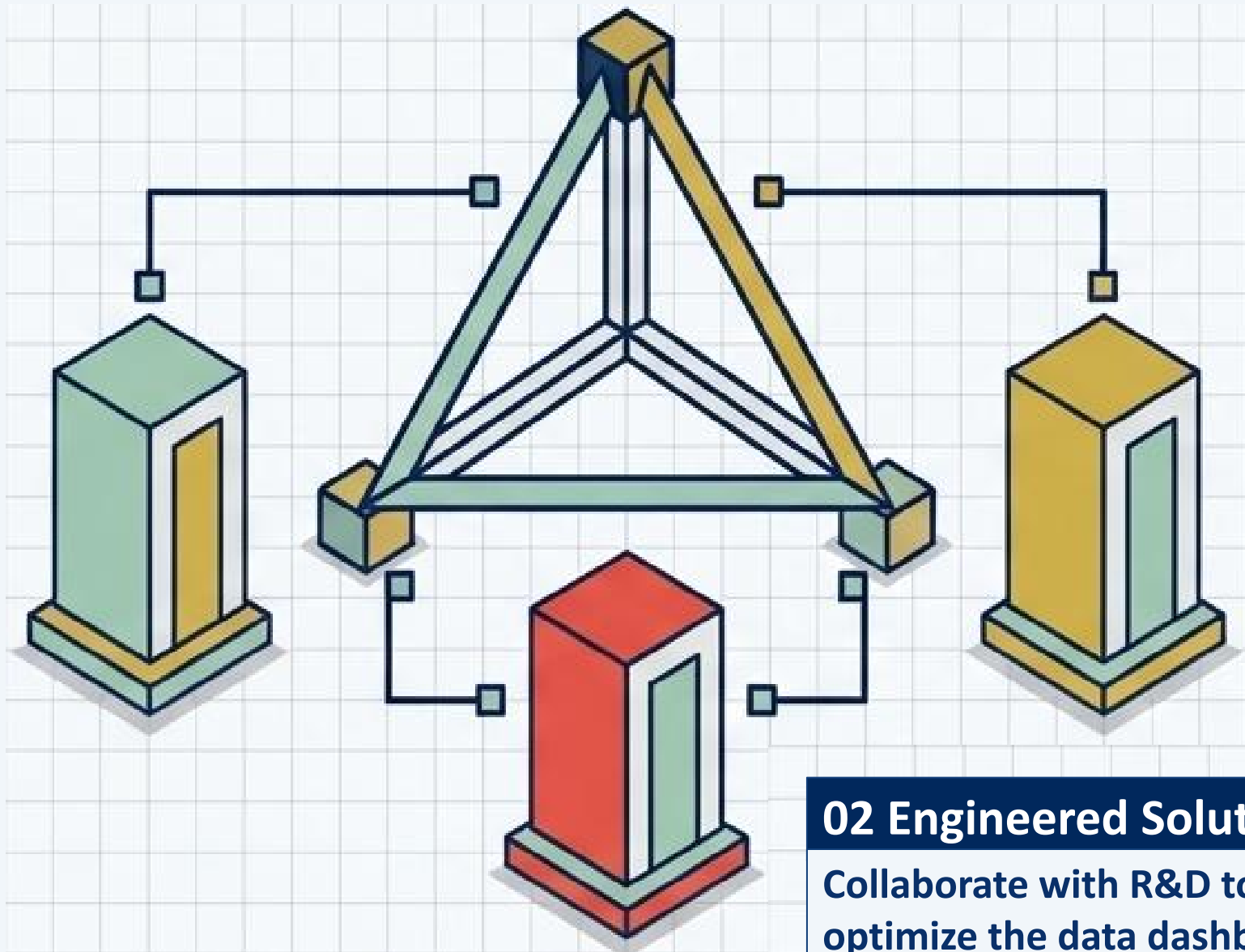
Conversion Funnel Analysis: Identifying GMV Bottlenecks via the VEE Model



Key Bottlenecks

1. Impression-to-click rate drop (**~88%**) driven by content-traffic relevance mismatch
2. Plunge in Click-to-Add-to-Cart Rate (**~78%**): Severe “price-value mismatch”
3. Payment → Repeat Purchase Plummet (**~72%**): Lack of a repeat purchase coupon mechanism

Key Recommendations: System-Level Architecture Upgrade Plan



01 The Flaw

The fixed threshold excludes high-quality small and medium-sized merchants, making it impossible to effectively increase GMV.

01 Engineered Solution

Dynamically adjust the eligibility thresholds for the funding pool

01 Target Impact

Merchant participation **increases by 15–20%.**
Goal: Funding pool fill rate **> 85%.**

03 The Flaw

Coordination gap between advertising placement and campaign activity budgets

03 Engineered Solution

Integrated Advertising and Campaigns Unified ROI Tracking

03 Target Impact

Resolve **internal KPI conflicts** and achieve **a win-win** outcome.

02 Engineered Solution

Collaborate with R&D to optimize the data dashboard

02 Target Impact

Response time has been reduced from **hours to minutes.**
Goal: Time from alert to action **<15 minutes.**

02 The Flaw

The T+1/T+2 data delay makes it impossible to intervene in a timely manner to prevent customer churn during the campaign period.

Key Conclusion:

Only by establishing a system-wide, cross-functional KPI framework can we eliminate conflicts at their root and transform siloed optimization into a cross-functional, system-wide growth strategy.

Critical Reflection: Limitations and Future Directions



Pain Point: Missing socio-cultural variables.

Over-reliance on internal platform perspectives blinds the system to local realities.



Future: Integrate anthropological data.



Pain Point:

Descriptive loops vs. lack of mathematical tipping points.

The existing R1/B1 feedback loops lack precision mathematical thresholds, making it impossible to accurately predict or preempt merchant churn.



Future: Deploy System Dynamics.



Pain Point: KPI-induced sub-optimization.

Internal friction and zero-sum games actively degrade performance.



Future: Platform Health Index.